Project 1: Reflection Paper

Ronald Nguyen, Alvin Reji, Jason Wong, Matthew Varela

Northeastern University

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Professor Cantilina

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The most difficult part in trying to articulate a solution to the problem presented was identifying the larger problem when it came to developing an animal enriching device. Our device needed to be able to tackle the problem long-term rather than provide temporary relief and entertainment to an intrinsic issue.

Following the decision of our animal, that being a cat, we looked at the largest struggles of cats. Speaking from personal experiences, some group members suggested tackling the predatory instincts of cats, which cause them to be overly aggressive and hostile at times. It is not uncommon to see cats constantly outside because they are not stimulated within the home and have the desire to hunt; however, this is extremely dangerous. There are many things in the outside world that put a cat's life at risk such as other predators or even humans. Our solution needed to be able to tame that aggression while simultaneously stimulating them in order to keep them from outside or just out of further danger. Entertaining a cat for that long posed enough of a challenge that needed a lot of planning and brainstorming of different ideas.

To start, we thought of the best incentives to provide for cats, which initially attracted us towards the idea of using food but ultimately led to us trying to find a safe way within the home to stimulate a cat's hunting senses. At first, we thought devices that involved food as an incentive were too generic, so we tried going down the path of devices that could imitate prey, such as a whack-a-mole machine or our initial prototype idea. This prototype involved multiple spools of string, which would be arranged to create a large, enclosed space, the string would be automatically revolving, and some type of cloth would be attached to the revolving string to imitate prey. However, this idea was shut down as we were unable to have the string revolve automatically without the use of electronic components or revolve efficiently through normal means.

We turned back to the idea of using food as an incentive in our device because although it is basic, it is efficient. We decided on creating a feeder machine, but with added levels of complexity by allowing cats to interact with the device to receive the food.

It was important that we took a good chunk of time designing the build of the device through the use of visuals and cardboard prototyping because we planned to have our final product fully 3D-printed. This takes a considerable amount of time, so we don't have many chances to go back and forth if certain parts of the project are dimensioned wrong or designed with large margins of error. We implemented both a spin mechanic and a push-pull mechanic in our device that would be easy for a cat to interact with.

We ran into our biggest problem in the design process with our first printed prototype. The outer cylinder was printed to exact dimensions creating a tight fit to the base that did not allow for our device to spin properly. However, we had a clear direction following this by improving the fit of the inner ring in order for it to spin with little friction while also being secured with a hollow fit at the top and bottom base. This was done by chamfering the edges of the bottom base and adding depth to the bases to extend beyond the rings. From here, our design process for the second layer and the rest of the device was straightforward, only requiring more thought to support the second layer and the entirety of the device to provide durability from the force of a cat. We added support beams within the first layer, which were held in by a friction fit rather than screws, and a bottom compartment that acts as a weight holder if needed to stabilize the device.

With each failed print, we had a reference to improve from our previous mistakes whether that was the functionality of the device or structure. One thing that was beneficial to us

was printing being at no cost to us due to our outside resources here on campus, but that just gave us more opportunity to improve and reflect on what was done wrong in previous prototypes.

Our final prototype proved to be very functional and relatively easy to use. One improvement that could be made would be making the device even more cat proof with locking mechanisms to limit the amount of food that is output or keep the device from being easily damaged. It could also be built upon with more layers for more complexity.

In all, our device provides cats with an innovative and entertaining way to retrieve food using a double-layered device that involves different mechanics and interactions that are stimulating and complex for a cat.